

SMS Validation Process Documentation

Overview

The SMS validation process is a comprehensive security and rate-limiting system designed to ensure secure SMS delivery while preventing abuse. The process consists of 7 sequential validation steps that verify various aspects of each SMS request before allowing it to be sent.

Validation Steps

Step 0: Initial Security Check

Purpose: Blocks high-risk requests before further processing.

Requirements:

- Requests must come directly (not through proxy connections)
- Requests must originate from 'example.com' domain unless coming from iOS platform

Failure Conditions:

- Proxy connection detected
- Non-iOS platform making request from unauthorized domain

Step 1: Google reCAPTCHA Validation

Purpose: Prevents automated/bot submissions on web platforms.

Process:

- Web platform requests must include valid Google reCAPTCHA token

- Token is verified through Google's reCAPTCHA service
- Non-web platforms skip this validation

Step 2: HTTP Origin Validation

Purpose: Ensures requests come from approved domains.

Approved Domains Example:

- admin.example.com
- example.com
- api.example.com
- partner.example.com

Validation Method:

- Extracts root domain from request origin header
- Compares against whitelist of approved domains

Step 3: Mobile Number Country Validation

Purpose: Restricts SMS sending to approved countries.

Key Rules:

- Mobile numbers must begin with country codes from approved list
- Special restriction for Oman (968) numbers from web platforms
- Bulk SMS requests skip country validation

Processing Steps:

1. Remove + sign and leading zeros from number

2. Extract first 3 digits as country code
3. Verify against allowed country codes list

Step 4: SMS Text Length Validation

Purpose: Ensures SMS content meets requirements.

Requirements:

- Minimum 1 character content
- Maximum 420 characters (3 SMS units)
- Does not apply to test server requests
- Recipient number not in blocked numbers list

Step 5: Rate Limiting Per Mobile Number

Purpose: Prevents SMS flooding to individual numbers.

Limit Rules:

- Maximum 1 SMS per 5 seconds per mobile number
- Uses sliding window tracking per number
- Blocks subsequent requests until time window passes

Step 6: Rate Limiting Per Source and Platform

Purpose: Enforces platform-specific sending limits.

Limit Structure:

- Separate per-minute and per-hour limits
- Different thresholds for iOS, Android, and web platforms

- Tracks counts per source (e.g., App/RegisterOTP)

Example Limits:

- iOS: 10/minute, 100/hour
- Web: 5/minute, 50/hour
- Android: 8/minute, 80/hour

Step 7: Logging and Sending SMS

Purpose: Final processing of validated requests.

Process Flow:

1. Create detailed audit log of SMS request including:
 - Source application
 - Recipient number
 - Message content
 - Request headers
 - Selected provider
2. Route SMS through configured provider (A, B, or C)
3. Return success/failure status to requester

Summary

The SMS validation process provides multiple layers of security and rate limiting to ensure legitimate use while preventing abuse. Each step addresses specific risks, from bot prevention to SMS flooding protection, with comprehensive logging for auditing purposes. The system maintains strict controls while allowing flexibility for different platforms and use cases.